

Entropy: Process Discovery Experiment Report

Compiled at 7:42am on Wednesday 10th December, 2025

1 Introduction

This report summarizes the setup and results of a process discovery experiment conducted using the *Entropy* tool. The following section outlines the discovery techniques used and their configurations in the experiment. Section 3 specifies the evaluation criteria used to assess the quality of the discovered models and to compare the performance of the discovery techniques. Finally, Section 4 presents the experimental results.

2 Discovery

The command line used to execute the experiment is listed below:

```
-mspd -el "BPI2012.xes" m1 -p 10 -mxItr 20 -t 5000 -mms 500 -pfs 200 -ebrm  
U - optm DFFA
```

Table 1 summarizes the key characteristics of the experimental setup used for performance comparison.

Table 1: Characteristics of the experiment

Parameter	Value
Population size	10
Iterations	20
Time limit (s)	5000
Pareto front size	200
Maximum model size	500
DCI grid siz	4
Optimized models	DFFA
Entropic relevance background model	U
Event log	BPI2012.xes

These discovery techniques are used in the experiment:

- DE [4];

3 Quality

The quality of the discovered models is assessed using two criteria: *entropic relevance*, which captures stochastic precision and recall [2], and *model size*, defined as the number of nodes and edges, which reflects model simplicity.

The performance of the discovery algorithms is evaluated using two aggregate measures: *dominance count* and the *Diversity Comparison Indicator* (DCI) [3]. Dominance count quantifies how many models discovered by a given algorithm dominate those from other algorithms in the global Pareto front. DCI measures the diversity of the discovered models by analyzing their distribution in the objective space. The details are provided in [1];

4 Results

Figure 1 presents the dominant discovered models with respect to entropic relevance and model size, where lower values are preferred for both criteria. Specifically, Fig. 1a, Fig. 1b, and Fig. 1c show the discovered deterministic frequency finite automata (DFFAs), stochastic deterministic finite automata (SDAGs), and directly-follows graphs (DFGs), respectively. The SDAG models are directly optimized, while the DFFA and DFG models are derived through translations of the optimized SDAGs. Table 5 in Appendix A lists the characteristics of all the discovered models.

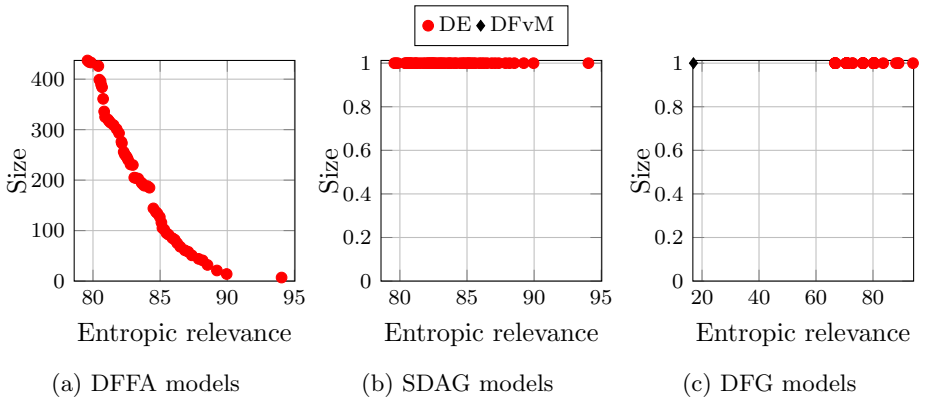


Figure 1: Discovered models plotted by size and entropic relevance for each representation type.

Table 2 and Table 3 summarize the analysis of the diversity of the models and the global Pareto front (dominance count), respectively. Table 4 reports the execution time (in iterations per second) for each input event log and method.

Table 2: Pareto dominance count comparison

Event log	DE(%)	Total
BPI2012.xes	100.00	8 100.00%

Table 3: DCI comparison (%)

Event log	DE(%)
BPI2012.xes	50.00

Table 4: Execution time comparison (iterations per second)

Event log	DE
BPI2012.xes	0.98

References

- [1] Hootan Zhian, Rajkumar Buyya, and Artem Polyvyanyy: *Multi-Objective Metaheuristics for Effective and Efficient Stochastic Process Discovery*. Proceedings of the 23rd International Conference on Business Process Management (BPM) Seville, Spain, August 31–September 5, 2025.
- [2] Hanan Alkhamash, Artem Polyvyanyy, Alistair Moffat, Luciano García-Bañuelos: *Entropic Relevance: A Mechanism for Measuring Stochastic Process Models Discovered From Event Data*. Inf. Syst. 107: 101922 (2022)
- [3] Miqing Li, Shengxiang Yang, Xiaohui Liu: *Diversity Comparison of Pareto Front Approximations in Many-Objective Optimization*. IEEE Trans. Cybern. 44(12): 2568–2584 (2014) Sander J.J. Leemans, Erik Poppe, Moe T. Wynn: *Directly Follows-Based Process Mining: Exploration & a Case Study*. In: ICPM, pp. 25–32 (2019)
- [4] Rainer Martin Storn, Kenneth Price: *Differential evolution - A simple and efficient adaptive scheme for global optimization over continuous spaces*. Tech. Rep. TR-95-012, International Computer Science Institute, 1947 Center Street, Berkeley (1995)

A Model Characteristics

Table 5 summarizes the quality characteristics of discovered models produced by the selected algorithms.

Table 5: Model characteristics: size and entropic relevance

Method	Type	Name	Visualization	Size	Entropic relevance
DE	DFFA	Candidate1	x	7.0	94.046
DE	DFFA	Candidate2	x	85.0	85.914
DE	DFFA	Candidate3	x	134.0	84.798
DE	DFFA	Candidate4	x	137.0	84.684
DE	DFFA	Candidate5	x	195.0	83.609
DE	DFFA	Candidate6	x	251.0	82.364
DE	DFFA	Candidate7	x	273.0	82.152
DE	DFFA	Candidate8	x	336.0	80.833
DE	DFFA	Candidate9	x	437.0	79.595
DE	DFFA	Candidate10	x	61.0	86.842
DE	DFFA	Candidate11	x	231.0	82.797
DE	DFFA	Candidate12	x	433.0	79.850
DE	DFFA	Candidate13	x	41.0	88.167
DE	DFFA	Candidate14	x	75.0	86.275
DE	DFFA	Candidate15	x	144.0	84.498
DE	DFFA	Candidate16	x	185.0	84.208
DE	DFFA	Candidate17	x	205.0	83.075
DE	DFFA	Candidate18	x	204.0	83.206
DE	DFFA	Candidate19	x	245.0	82.536
DE	DFFA	Candidate20	x	293.0	81.940
DE	DFFA	Candidate21	x	361.0	80.755
DE	DFFA	Candidate22	x	68.0	86.498
DE	DFFA	Candidate23	x	314.0	81.302
DE	DFFA	Candidate24	x	44.0	87.867
DE	DFFA	Candidate25	x	32.0	88.529
DE	DFFA	Candidate26	x	51.0	87.363
DE	DFFA	Candidate27	x	92.0	85.627
DE	DFFA	Candidate28	x	188.0	84.058
DE	DFFA	Candidate29	x	230.0	82.969
DE	DFFA	Candidate30	x	301.0	81.763
DE	DFFA	Candidate31	x	95.0	85.472
DE	DFFA	Candidate32	x	14.0	89.956
DE	DFFA	Candidate33	x	105.0	85.172
DE	DFFA	Candidate34	x	21.0	89.230
DE	DFFA	Candidate35	x	203.0	83.372
DE	DFFA	Candidate36	x	320.0	81.121
DE	DFFA	Candidate37	x	82.0	86.089
DE	DFFA	Candidate38	x	317.0	81.198
DE	DFFA	Candidate39	x	248.0	82.455
DE	DFFA	Candidate40	x	102.0	85.310
DE	DFFA	Candidate41	x	240.0	82.619
DE	DFFA	Candidate42	x	309.0	81.535
DE	DFFA	Candidate43	x	192.0	83.702

Method	Type	Name	Visualization	Size	Entropic relevance
DE	DFFA	Candidate44	x	256.0	82.287
DE	DFFA	Candidate45	x	426.0	80.399
DE	DFFA	Candidate46	x	127.0	84.976
DE	DFFA	Candidate47	x	399.0	80.477
DE	DFFA	Candidate48	x	189.0	83.822
DE	DFFA	Candidate49	x	384.0	80.677
DE	DFFA	Candidate50	x	276.0	82.118
DE	DFFA	Candidate51	x	434.0	79.716
DE	DFFA	Candidate52	x	58.0	87.093
DE	DFFA	Candidate53	x	325.0	80.902
DE	DFFA	Candidate54	x	389.0	80.608
DE	DFFA	Candidate55	x	116.0	85.095
DE	DFFA	Candidate56	x	396.0	80.563
DE	SDAG	Candidate1	x	1.0	94.046
DE	SDAG	Candidate2	x	1.0	85.913
DE	SDAG	Candidate3	x	1.0	84.794
DE	SDAG	Candidate4	x	1.0	84.680
DE	SDAG	Candidate5	x	1.0	83.596
DE	SDAG	Candidate6	x	1.0	82.349
DE	SDAG	Candidate7	x	1.0	82.136
DE	SDAG	Candidate8	x	1.0	80.816
DE	SDAG	Candidate9	x	1.0	79.588
DE	SDAG	Candidate10	x	1.0	86.841
DE	SDAG	Candidate11	x	1.0	82.783
DE	SDAG	Candidate12	x	1.0	79.842
DE	SDAG	Candidate13	x	1.0	88.167
DE	SDAG	Candidate14	x	1.0	86.273
DE	SDAG	Candidate15	x	1.0	84.493
DE	SDAG	Candidate16	x	1.0	84.199
DE	SDAG	Candidate17	x	1.0	83.061
DE	SDAG	Candidate18	x	1.0	83.193
DE	SDAG	Candidate19	x	1.0	82.524
DE	SDAG	Candidate20	x	1.0	81.923
DE	SDAG	Candidate21	x	1.0	80.738
DE	SDAG	Candidate22	x	1.0	86.498
DE	SDAG	Candidate23	x	1.0	81.284
DE	SDAG	Candidate24	x	1.0	87.867
DE	SDAG	Candidate25	x	1.0	88.529
DE	SDAG	Candidate26	x	1.0	87.363
DE	SDAG	Candidate27	x	1.0	85.626
DE	SDAG	Candidate28	x	1.0	84.048
DE	SDAG	Candidate29	x	1.0	82.956
DE	SDAG	Candidate30	x	1.0	81.744
DE	SDAG	Candidate31	x	1.0	85.471

Method	Type	Name	Visualization	Size	Entropic relevance
DE	SDAG	Candidate32	x	1.0	89.956
DE	SDAG	Candidate33	x	1.0	85.169
DE	SDAG	Candidate34	x	1.0	89.230
DE	SDAG	Candidate35	x	1.0	83.359
DE	SDAG	Candidate36	x	1.0	81.103
DE	SDAG	Candidate37	x	1.0	86.088
DE	SDAG	Candidate38	x	1.0	81.181
DE	SDAG	Candidate39	x	1.0	82.442
DE	SDAG	Candidate40	x	1.0	85.307
DE	SDAG	Candidate41	x	1.0	82.607
DE	SDAG	Candidate42	x	1.0	81.516
DE	SDAG	Candidate43	x	1.0	83.690
DE	SDAG	Candidate44	x	1.0	82.272
DE	SDAG	Candidate45	x	1.0	80.380
DE	SDAG	Candidate46	x	1.0	84.973
DE	SDAG	Candidate47	x	1.0	80.459
DE	SDAG	Candidate48	x	1.0	83.812
DE	SDAG	Candidate49	x	1.0	80.660
DE	SDAG	Candidate50	x	1.0	82.101
DE	SDAG	Candidate51	x	1.0	79.708
DE	SDAG	Candidate52	x	1.0	87.093
DE	SDAG	Candidate53	x	1.0	80.886
DE	SDAG	Candidate54	x	1.0	80.590
DE	SDAG	Candidate55	x	1.0	85.091
DE	SDAG	Candidate56	x	1.0	80.545
DE	DFG	Candidate1	x	1.0	94.046
DE	DFG	Candidate2	x	1.0	80.384
DE	DFG	Candidate3	x	1.0	80.242
DE	DFG	Candidate4	x	1.0	80.242
DE	DFG	Candidate5	x	1.0	76.483
DE	DFG	Candidate6	x	1.0	70.878
DE	DFG	Candidate7	x	1.0	70.878
DE	DFG	Candidate8	x	1.0	70.702
DE	DFG	Candidate9	x	1.0	66.752
DE	DFG	Candidate10	x	1.0	80.384
DE	DFG	Candidate11	x	1.0	72.826
DE	DFG	Candidate12	x	1.0	66.752
DE	DFG	Candidate13	x	1.0	83.582
DE	DFG	Candidate14	x	1.0	80.384
DE	DFG	Candidate15	x	1.0	80.242
DE	DFG	Candidate16	x	1.0	76.483
DE	DFG	Candidate17	x	1.0	76.483
DE	DFG	Candidate18	x	1.0	76.483
DE	DFG	Candidate19	x	1.0	70.878

Method	Type	Name	Visualization	Size	Entropic relevance
DE	DFG	Candidate20	x	1.0	70.702
DE	DFG	Candidate21	x	1.0	70.702
DE	DFG	Candidate22	x	1.0	80.384
DE	DFG	Candidate23	x	1.0	70.702
DE	DFG	Candidate24	x	1.0	80.384
DE	DFG	Candidate25	x	1.0	88.079
DE	DFG	Candidate26	x	1.0	80.384
DE	DFG	Candidate27	x	1.0	80.384
DE	DFG	Candidate28	x	1.0	76.483
DE	DFG	Candidate29	x	1.0	72.826
DE	DFG	Candidate30	x	1.0	70.702
DE	DFG	Candidate31	x	1.0	80.384
DE	DFG	Candidate32	x	1.0	88.936
DE	DFG	Candidate33	x	1.0	80.384
DE	DFG	Candidate34	x	1.0	88.936
DE	DFG	Candidate35	x	1.0	76.483
DE	DFG	Candidate36	x	1.0	70.702
DE	DFG	Candidate37	x	1.0	80.384
DE	DFG	Candidate38	x	1.0	70.702
DE	DFG	Candidate39	x	1.0	70.878
DE	DFG	Candidate40	x	1.0	80.384
DE	DFG	Candidate41	x	1.0	70.878
DE	DFG	Candidate42	x	1.0	70.702
DE	DFG	Candidate43	x	1.0	76.483
DE	DFG	Candidate44	x	1.0	70.878
DE	DFG	Candidate45	x	1.0	66.752
DE	DFG	Candidate46	x	1.0	80.242
DE	DFG	Candidate47	x	1.0	66.752
DE	DFG	Candidate48	x	1.0	76.483
DE	DFG	Candidate49	x	1.0	66.752
DE	DFG	Candidate50	x	1.0	70.702
DE	DFG	Candidate51	x	1.0	66.752
DE	DFG	Candidate52	x	1.0	80.384
DE	DFG	Candidate53	x	1.0	70.702
DE	DFG	Candidate54	x	1.0	66.752
DE	DFG	Candidate55	x	1.0	80.242
DE	DFG	Candidate56	x	1.0	66.752